

PSID: 1603		Project Summary Report		Team ID: SIH123	
Title		Research and develop a design on (autonomous water surface cleaning robot)		Theme software	
Objective		To design <u>a</u> autonomous robot for surface cleaning of water bodies and research about the methods of inclusion of AI and IOT			
—Synopsis (Background & Purpose)		Using fusion360 to design and optimize the parts used in the autonomous water surface cleaning robot and to include AI and <u>iot</u> for novelty and efficiency			
Methodology (Flow Chart, Process Chart, etc.) Along with Real Time Product Picture		The methodology is attached herewith.			
Products (Workshops, materials, skills developed) & Product Availability		The materials used vary from the range of metals to plastic with the use of lightweight aluminium metal body and the pontoons of high density polyethene and <u>polypropylene</u> . The product details and cost considerations are attached herewith.			
Outcomes & Future Plan		The future plan includes to reduce production costs and improvising on the capacity of trash college and a better method to involve communities for cleanliness in a <u>cheaper methods</u> .			

PRAYAS

Sanjog Panda

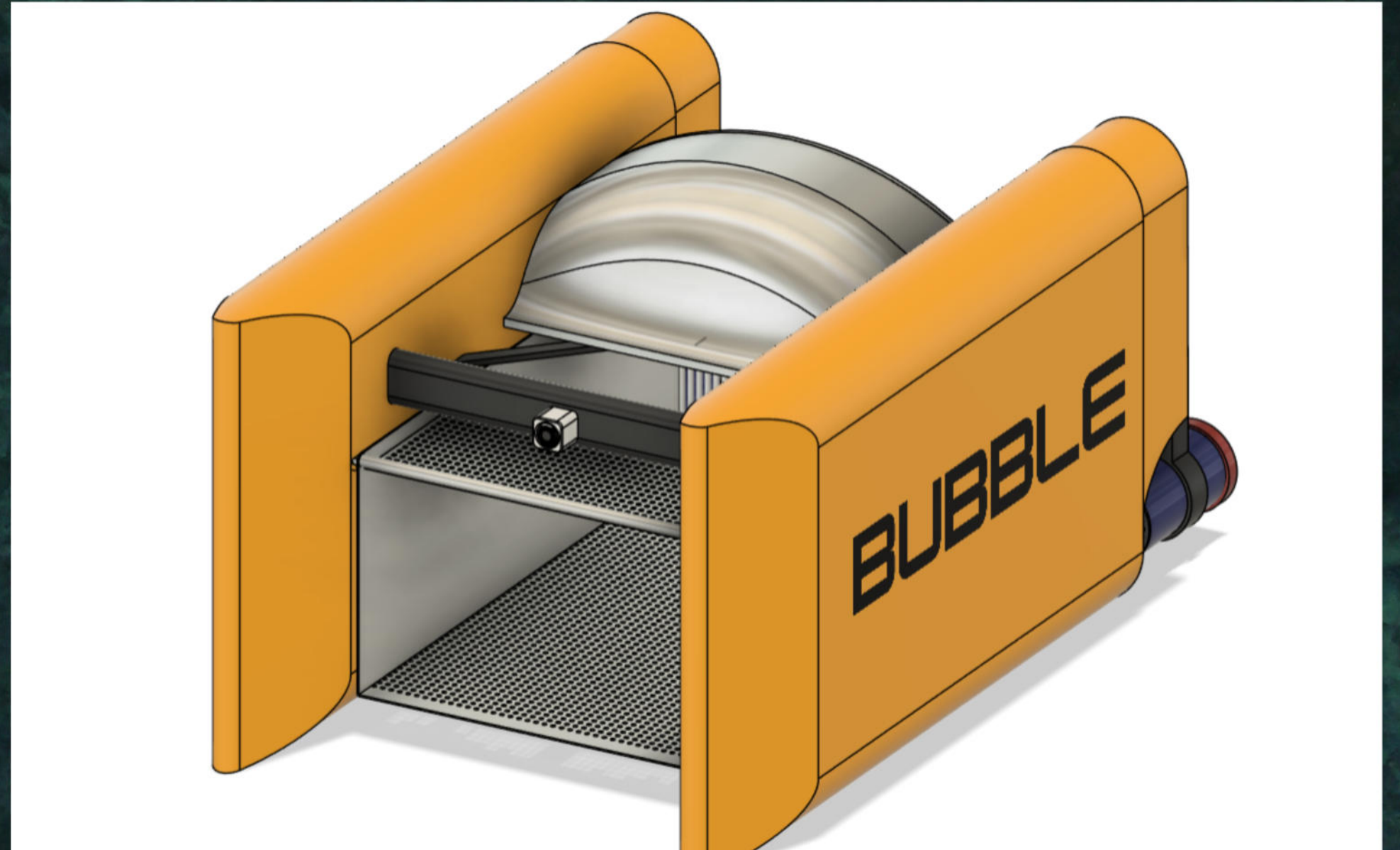
Daksh Goswami

M. Sameel

Surya prabha

Abdul Raheem

Aashish Gautam



Autonomous Water Surface Cleaning Robot

Problem Statement

To design an autonomous water surface cleaning robot using fusion 360 and to optimize the components using generative design and research use of IoT and AI along with it

Design features

- Integration of AI and IoT
- Light weight floating body
- Marine grade aluminium trash cage
- Oil absorbing Mechanism

SENSORS

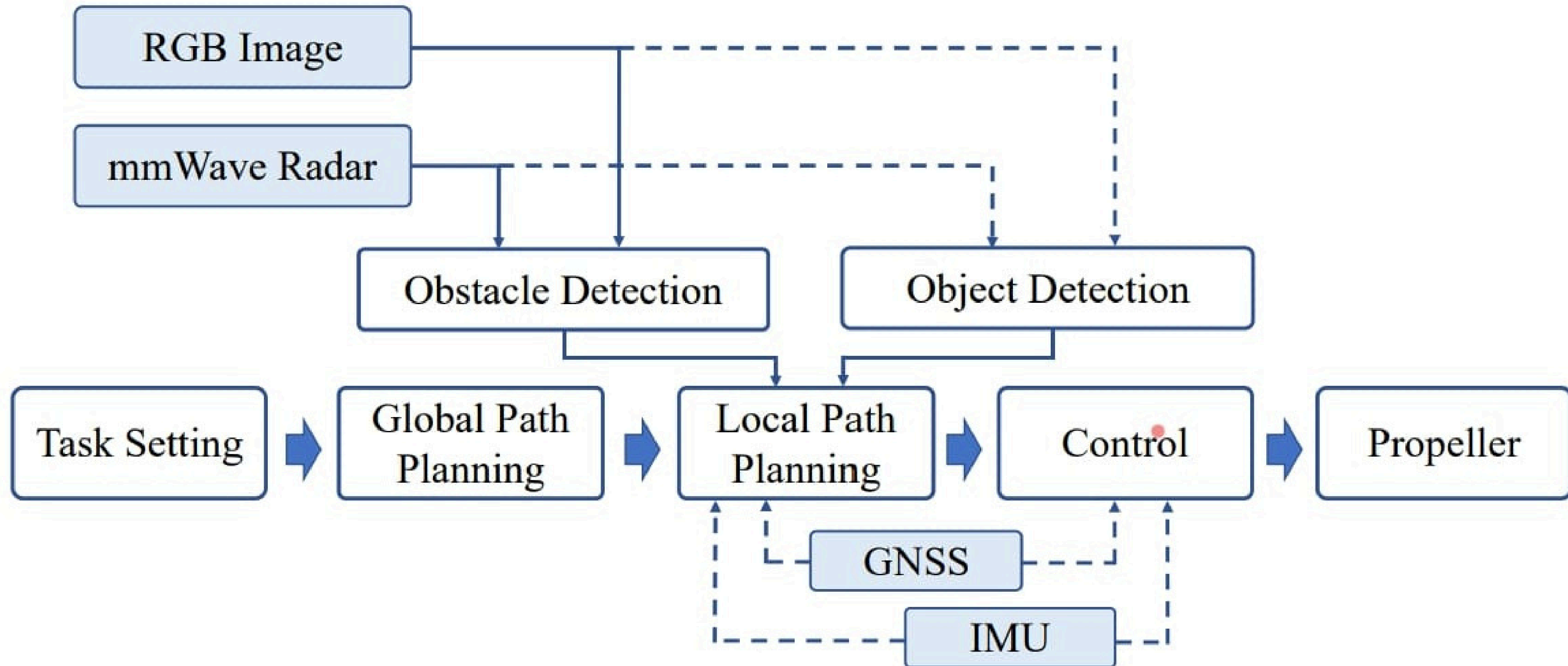
NVIDIA Xavier NX Module

IMU and RCB GNSS

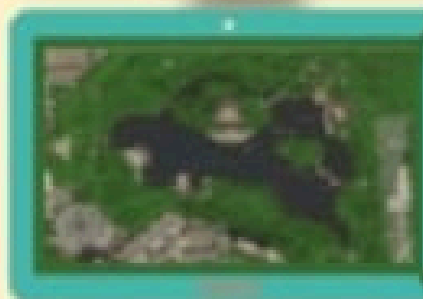
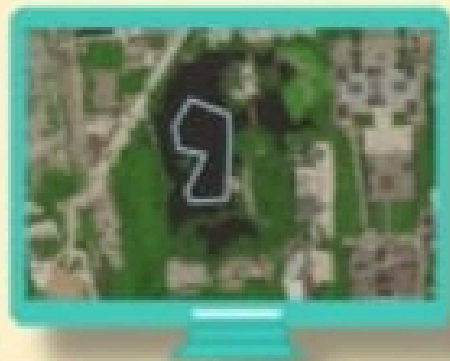
mm Wave Radar

Onboard RGB camera

Workflow



Initialization



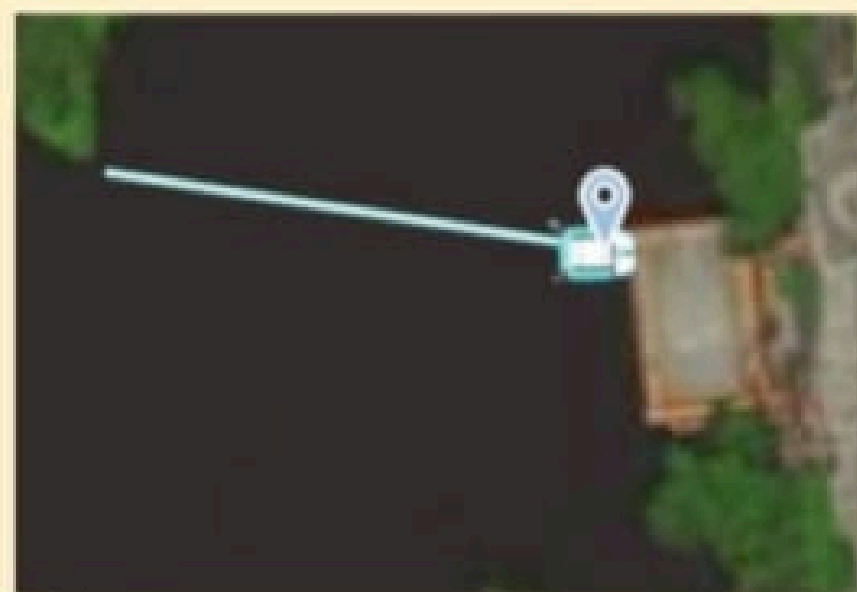
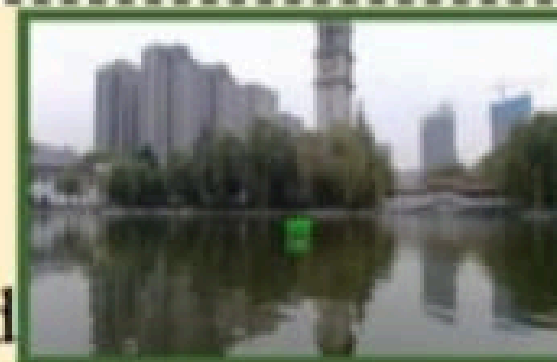
Initialization result upload
through 4G Internet



Mobile phone, Pad, or PC Set the cleaning boundary and a home point

Autonomous Task Performing

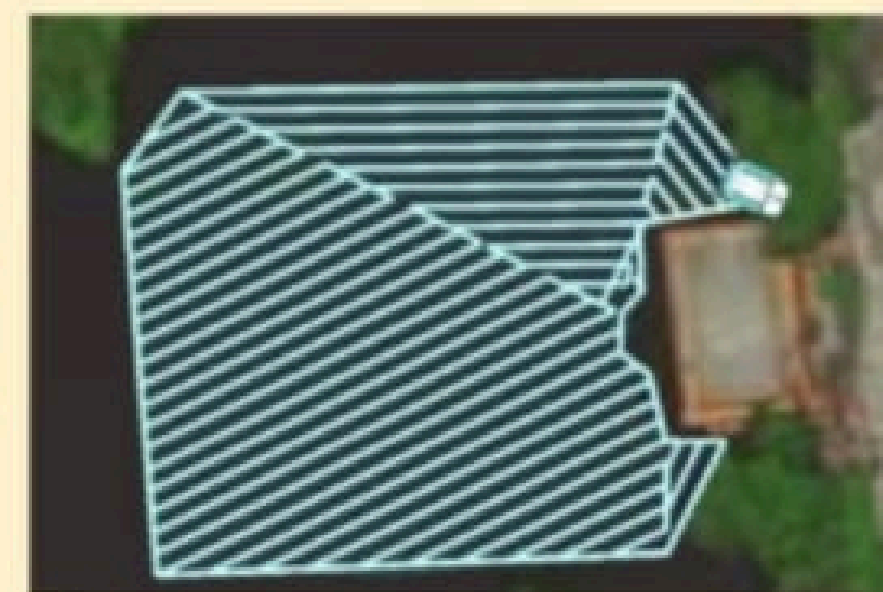
Switch to the local path
when wastes are detected



Back to home point



Sails along the global path



Global path planning

USE OF CPP(Coverage Path Planning) ALGORITHM



(a)



(b)



(c)



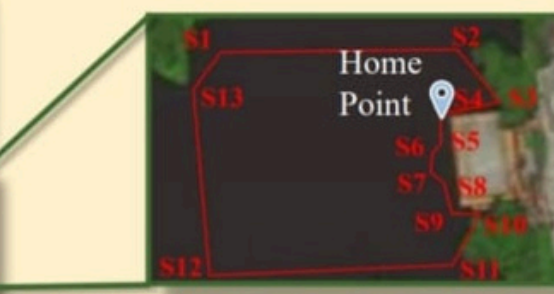
(d)



Initialization



Mobile phone, Pad, or PC



Set the cleaning boundary and a home point

Initialization result upload through 4G Internet

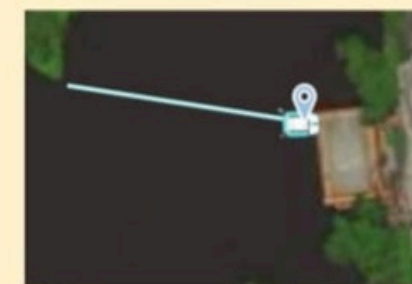


Cloud

Load to SMURF through 4G Internet

Autonomous Task Performing

Switch to the local path when wastes are detected



Back to home point



Sails along the global path



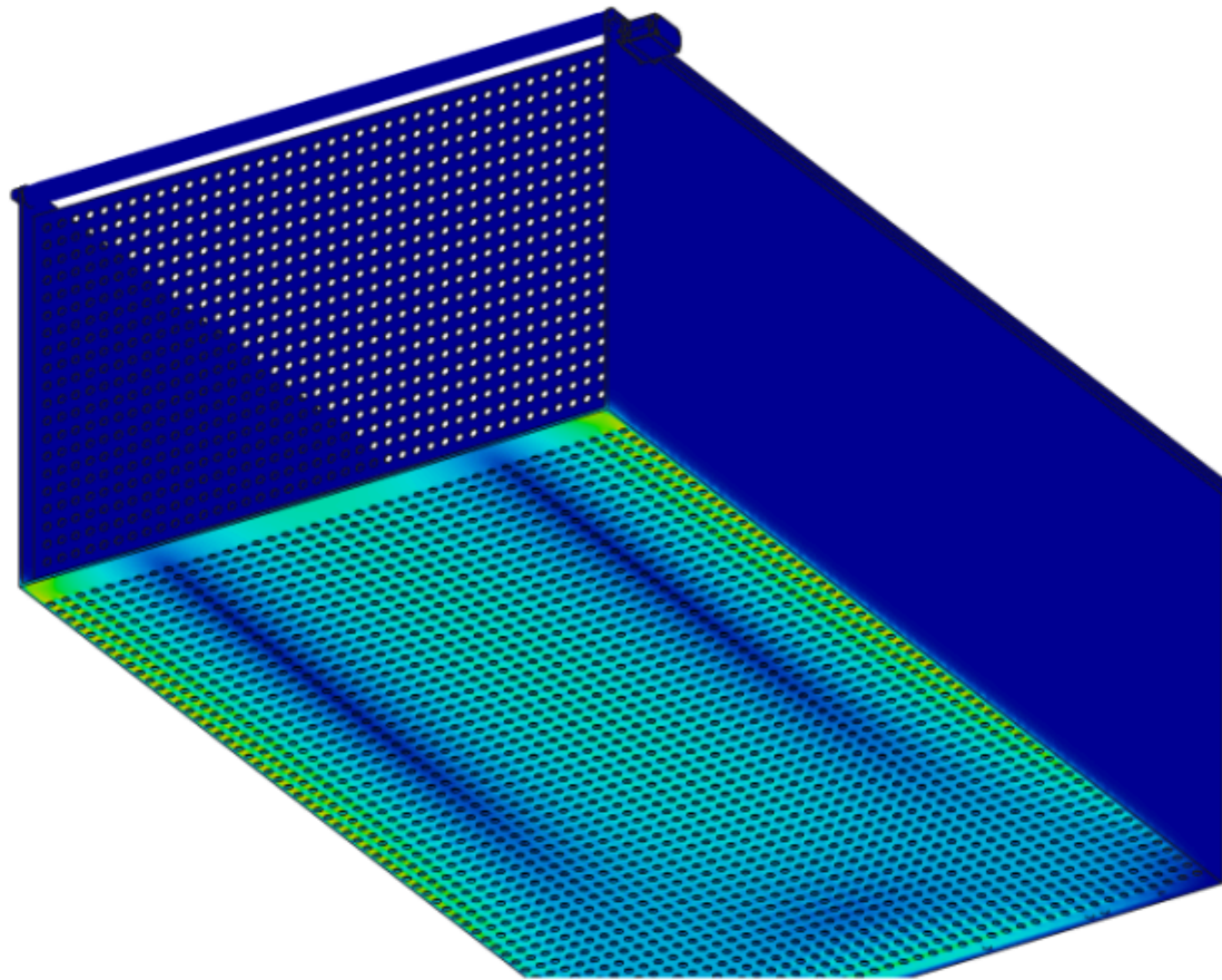
Global path planning

Static stress simulation on Trash Basket

▣ **Stress**

▣ **von Mises**

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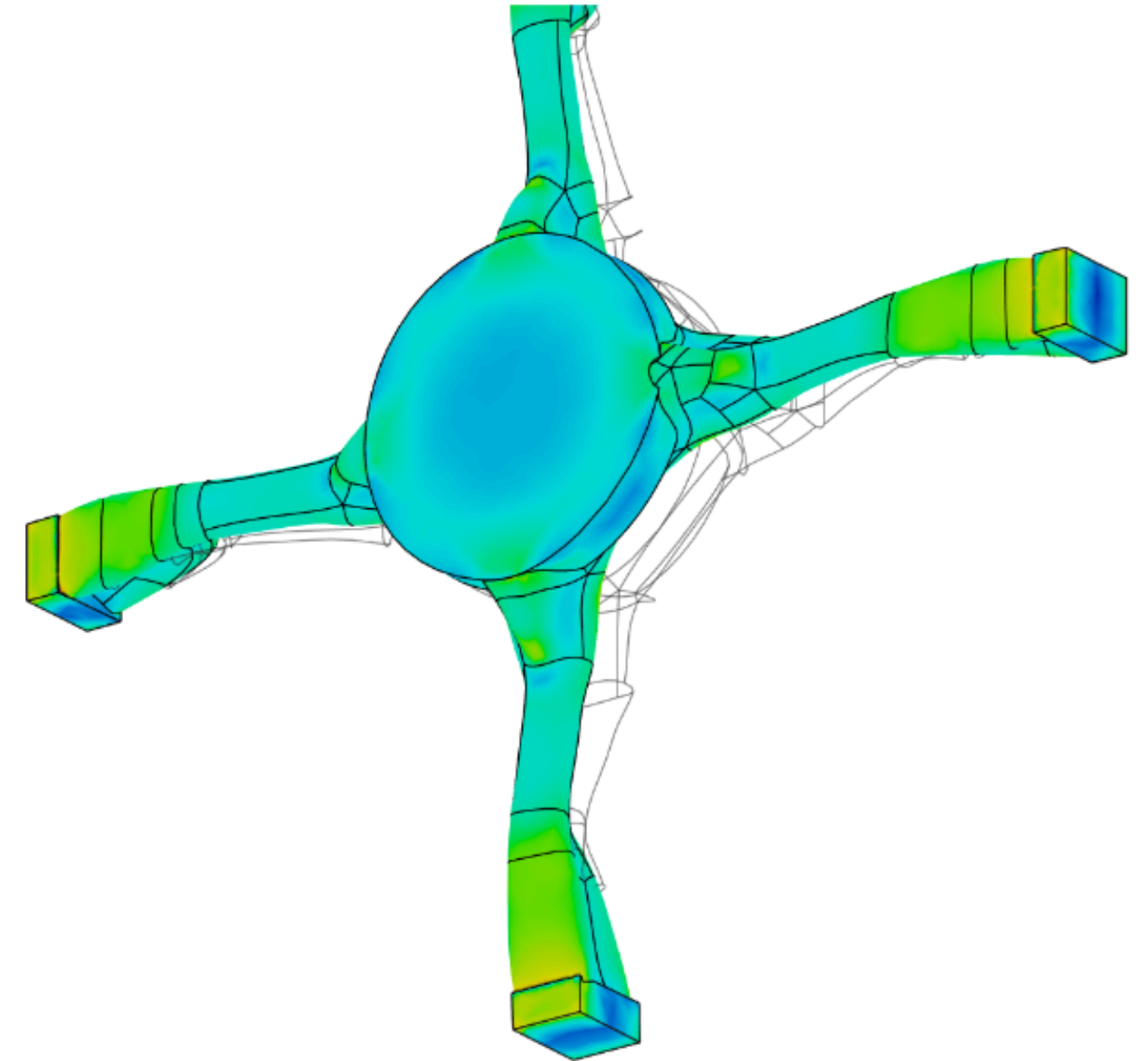


stress analysis of the trash cage
<https://a360.co/3Tu8pxl>

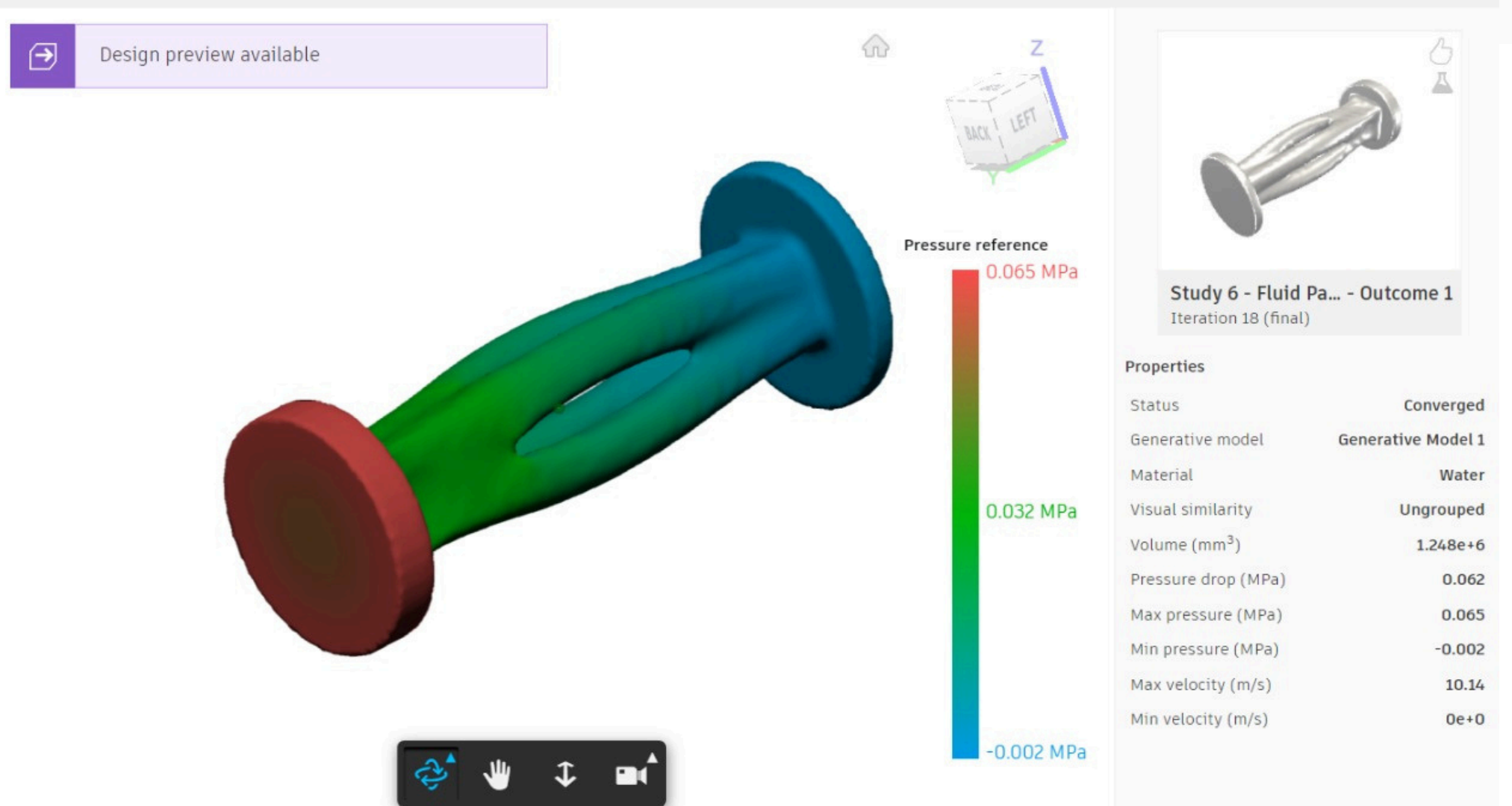
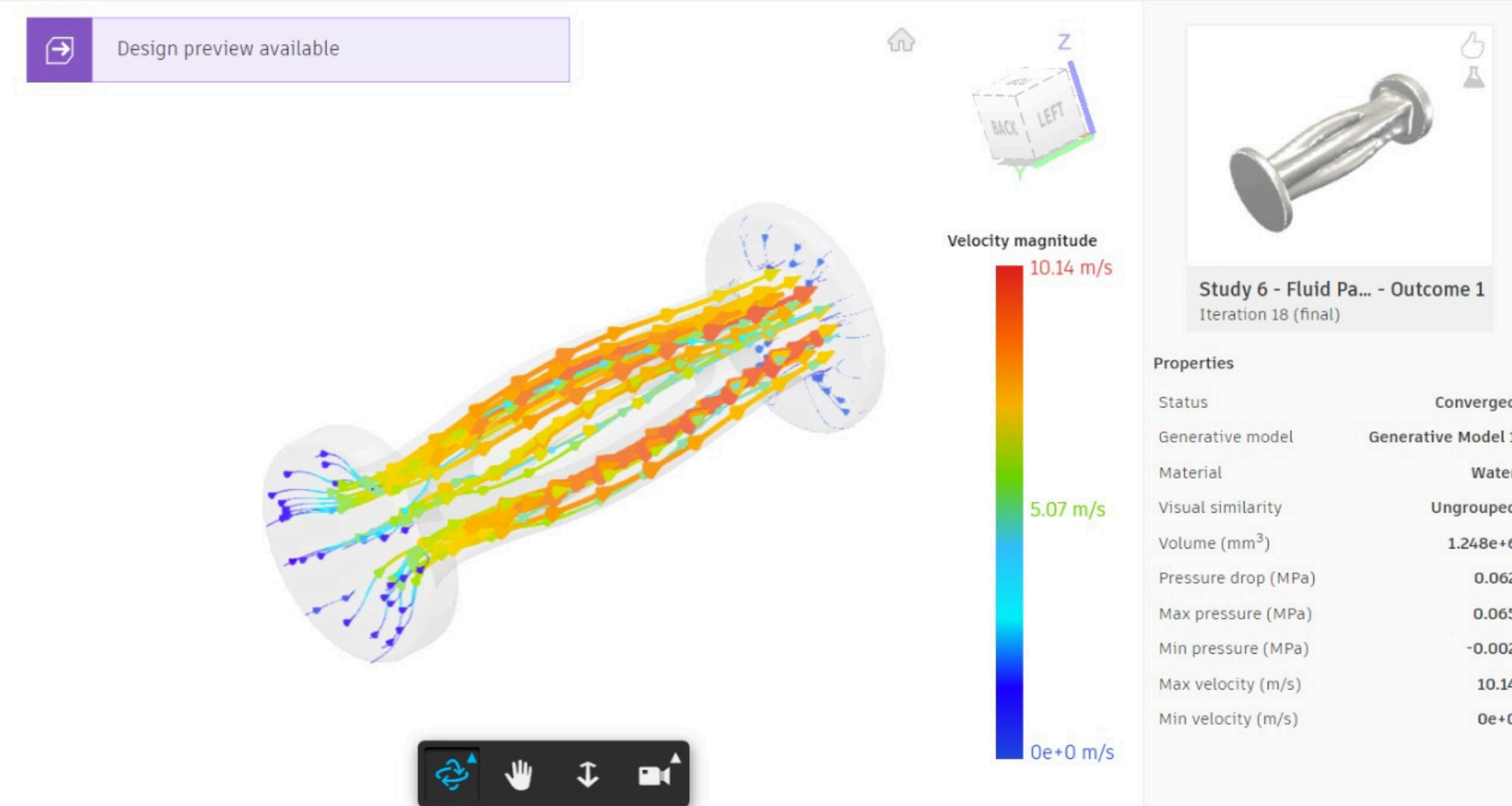
Generative Design On Motor Mount

▣ **von Mises**

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generative design optimization
<https://a360.co/3ZsLohW>



Fluid flow simulation

Stress Analysis of the Trash Cage

<https://a360.co/3Tu8pxl>



Generative Design Optimization:

<https://a360.co/3ZsLohW>

Estimation(electronics)

components	cost	dimension/weight
battery	15,000(2*7,500)	18*16*7/4kg
Nvidia xavier <u>nx</u> chip	50,000	10*5*2/0.18kg
Esp 32	500	5*3*1
servo	5,000	4*4*2
Rgb camera	1,500	3*3*3
IMU RTK-GNSS	23,000	1.6*6
MM wave radar	2,500	11*7
Propeller motors	44,000(2) 	30*13.5

Cost Estimation(Hardware)

components	weight	dimension	material	cost
Hull or pontoon	5-15kg	150*100*75	HDPE	70/kg
Central cage	15 kg	100*60*40	Marine grade aluminium 6061	300-500/kg
Propeller Propulsion cage	3-5kg	80mm 30*13.5	Plastic	22,000(with motor)
Cage shutter	2-3 kg		aluminium	300-500/kg
Oil collector	-	Sheets 2mm	Polyurethane foam	200/sheet

Performance of the robot

3 hours continuous action

Easy deployment and trash collection

Emergency troubleshooting

Efficient AI algorithms

Realtime area mapping and processing

Efficient in Harsh Conditions

Future enhancements and challenges

- Cost Reduction
- Technical Advancements
- Capacity of Trash Collector
- Advanced Pollution Sensors